

In This Issue

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This special section of EMSE offers extended versions of three papers selected from the 2004 International Symposium on Software Reliability Engineering (ISSRE). ISSRE is sponsored by the IEEE Reliability Society and IEEE Computer Society and provides a venue for dissemination of state-of-the-art research and practice in the engineering of dependable software systems. Empirical methods play a central role in software dependability engineering. The empirical investigation of the dependability of deployed systems requires methods for predicting and increasing their tolerance to failures. Rigorous empirical evaluation processes can provide faithful data for critical quality attributes of complex software systems, such as reliability, availability, and security. Thus, a special section of EMSE is devoted to the best papers from ISSRE 2004.

The papers which are presented here are the result of a rigorous selection process. Of the 120 submissions 39 papers were selected for inclusion in the Proceedings. For this special issue, we identified five high-quality papers, with a focus on empirical studies. The authors were invited to submit an extended version of their ISSRE 2004 paper, and the revised papers then underwent a second review process, with three papers chosen for this special section of EMSE.

The first paper titled “The Economics of Unit Testing” by Ellims, Bridges and Ince focuses on unit testing, often criticized for the amount of time and cost required. However the results presented in this paper, obtained from three safety-related, industrial software projects, provide evidence that the perceived costs of unit testing may be exaggerated and that the likely benefits in terms of defect detection are quite high in relation to those costs. Particular attention is devoted to a discussion of the results obtained at different phases of development using different methods to generate the tests and to the comparison with empirical results in the literature.

The second paper titled “Prioritizing JUnit Test Cases: An Empirical Assessment and Cost-Benefits Analysis” by Do, Rothermel, and Kinneer represents a first attempt to expand the investigation of test case prioritization to Java and the JUnit testing framework. Initial investigations focused on a single procedural language, C, with a limited set of test suites. This paper extends and compares previous results by performing a controlled experiment on Java programs. It also demonstrates that test case prioritization can significantly improve the rate of fault detection of JUnit test suites and analyzes how the language and testing paradigm can influence the results obtained.

The third paper titled “Empirical Characterization of Session-Based Workload and Reliability for Web Servers” by Goseva-Popstojanova, Singh, Mazimdar, and Li focuses on the availability of Internet access, by presenting a detailed empirical

analysis of the session-based workload and reliability, based on the data extracted from actual Web logs of eleven Web servers. For this purpose, several intra-session and inter-session metrics are defined that collectively describe Web workload in terms of user sessions, and estimate Web servers' request-based and session-based reliability. The results presented show that session-based workload and reliability are better indicators of the users' perception of Web quality than the request-based metrics.

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Eda Marchetti

Guest Editors for the special issue on ISSRE 2004 best papers

In addition to the special issue described above, we have two regular papers. The first paper, by Maldonado et al, entitled "Perspective-Based Reading: A Replicated Experiment Focused on Individual Reviewer Effectiveness," describes a replication conducted to compare the effectiveness of inspectors using Perspective Based Reading (PBR) to the effectiveness of inspectors using a checklist. The goal of this replication was to better understand how complementary are PBR perspectives. Results show that PBR was more effective than Checklist for one of the two requirements documents used. The results also showed that for the same document on which PBR was more effective, the PBR and Checklist techniques were complementary. Perhaps more importantly, in most cases the perspectives turned out to find different defects (i.e., they were not correlated), in the case where the perspectives found similar defects (i.e., correlated) there was no overall benefit observed for the perspective-based technique.

The second paper, by Syed-Abdullah, Holcombe, and Gheorge, entitled "The Impact of an Agile Methodology on the Well Being of Development Teams," describes an empirical study, which addresses the question of whether an agile methodology has any effect on the well being of the software developers. A collection of quantitative and qualitative methods were used, including participative observation, focus group interviews, close-ended questionnaires, Spearman Correlation, and the Mann-Whitney test. Initial results showed that the effect of an agile methodology (XP) on developers depends on the types of projects that are undertaken, i.e., in general, the well being of the developers seemed to be better only when the requirements are uncertain or volatile.

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Editors-in-Chief